

EDUCATION

Southern University of Science and Technology

Sep. 2023 – Jun. 2026

*M.phil. in Computer Science — Advised by Prof. Bo Tang**ShenZhen, China*

- GPA: 3.62/4.00 (90/100)

Southern University of Science and Technology

Sep. 2019 – Jun. 2023

*B.Eng. in Computer Science — Advised by Prof. Bo Tang & Prof. Xiao Yan**ShenZhen, China*

- GPA: 3.82/4.00 (91/100)
- Key Coursework: Linear Algebra (93), Discrete Math (96), Algorithm Design & Analysis (97), Database Principles (91), Computer Organization (94), Computer Network (96), Artificial Intelligence (92)
- Language Proficiency: TOEFL (104), CET-6 (590)

EXPERIENCE

NYU System Group, New York University

Dec. 2024 – Jun. 2025

Student Intern — Advised by Prof. Jinyang Li

- General and effective **GraphRAG** solutions and robust **Agent** program development.

Shanghai AI Lab, Amazon Web Services

Jun. 2022 – Nov. 2024

Applied Scientist Intern in DGL Team — Advised by Dr. Minjie Wang

- Efficient **GNN training** on large-scale datasets in single-machine and distributed environments.

PUBLICATIONS

* **PolyG: Effective and Efficient GraphRAG with Adaptive Graph Traversal**

Renjie Liu, Haitian Jiang, Xiao Yan, Bo Tang, and Jinyang Li

<https://arxiv.org/pdf/2504.02112>* **DiskGNN: Bridging I/O Efficiency and Model Accuracy for Out-of-Core GNN Training**

Renjie Liu*, Yichuan Wang*, Xiao Yan, Zhenkun Cai, Minjie Wang, Haitian Jiang, Bo Tang, and Jinyang Li

ACM Conference on Management of Data (SIGMOD 2025)

*Equal Contribution

* **APT: Adaptive Parallel Training for Graph Neural Networks**

Kaihao Ma*, Renjie Liu*, Zhenkun Cai, Xiao Yan, Xiang Song, Minjie Wang, and James Cheng

ACM Symposium on Principles and Practice of Parallel Programming (PPoPP 2025)

*Equal Contribution

* **Forming Scalable, Convergent GNN Layers that Minimize a Sampling-Based Energy**

Haitian Jiang, Renjie Liu, Xiao Yan, Zengfeng Huang, Zhenkun Cai, Minjie Wang, and David Wipf

*International Conference on Learning Representations (ICLR 2025)** **gSampler: General and Efficient GPU-based Graph Sampling for Graph Learning**

Ping Gong, Renjie Liu, Zunyao Mao, Zhenkun Cai, Xiao Yan, Cheng Li, Minjie Wang, and Zhuozhao Li

ACM Symposium on Operating Systems Principles (SOSP 2023)

TECHNICAL SKILLS

Programming Languages: Python (system abstraction), C/C++ (core operators), CUDA (GPU kernels).**Frameworks and Tools:** vLLM, Pytorch, DGL, PyG, Thrust, SQL, Latex, Git, etc.

RESEARCH PROJECTS

Effective and Efficient Graph RAG Solution (PolyG)

Aug. 2024 – May 2025

- Construct a new benchmark tailored for Graph RAG queries and propose a new approach that is both effective and efficient with adaptive graph traversal under diverse query patterns.
 - ⇒ Observe that existing GraphQA benchmarks lack query diversity, and create the first Graph RAG benchmark.
 - ⇒ Propose a Graph RAG solution that adaptively selects the best traversal methods based on the query pattern.

Efficient out-of-core GNN Training on Large-scale Graphs (DiskGNN) Sep. 2023 – July 2024

- Build a system which achieves high I/O efficiency and thus fast training without hurting model accuracy for out-of-core GNN training where the whole graph can not fit in CPU memory.
 - ⇒ Observing that offline sampling scheme does not degrade model accuracy by sampling mini-batches beforehand.
 - ⇒ Propose a suite of designs based on offline sampling, including four-level feature store and batched feature packing.

Adaptive Parallel Training for Graph Neural Networks (APT)

April 2023 – Dec. 2023

- Build a general framework capable of automatically determine the best parallelism for distributed GNN training based on hardware environments and models.
 - ⇒ Derive an abstraction for all parallel schemes from the perspective of matrix segmentation & multiplication.
 - ⇒ Design a cost model to automatically determine the best parallelism given specific environments and configurations.

General and Efficient GPU-based Graph Sampling (gSampler)

Aug. 2022 – April 2023

- Build a high-performance GPU sampling framework for GNN training that can express diverse sampling algorithms by sketching sampling process using matrix operations.
 - ⇒ Propose matrix-centric APIs based on a proposed 4-step ECSF model to sketch the graph sampling process.
 - ⇒ Incorporate a suite of sampling-oriented optimizations, e.g., kernel fusion, format selection and batching.

HONORS AND AWARDS

- China National Scholarship Nov. 2024
- University Merit Scholarship for Outstanding Student: The 1st Class Sep. 2023, 2022
- Outstanding Graduates of Department of Computer Science and Engineering Jun. 2023
- ACM SIGMOD 2022 Programming Contest: World's Finalist April 2022

TEACHING EXPERIENCE

Teaching Assistant

- **Computer Organization and Architecture** Mar. 2022 – Jun. 2022
 - Designed and graded course assignments and the course project (a Minisys CPU written in Verilog on FPGA).
 - Answered questions about CPU architecture, MIPS and Verilog programming in laboratory courses.
- **Introduction to Computer Programming** Sep. 2020 – Jun. 2021
 - Designed and graded course assignments and the course project (a Minesweeper implemented by Java).
 - Answered questions about programming skills and algorithms in laboratory classes.